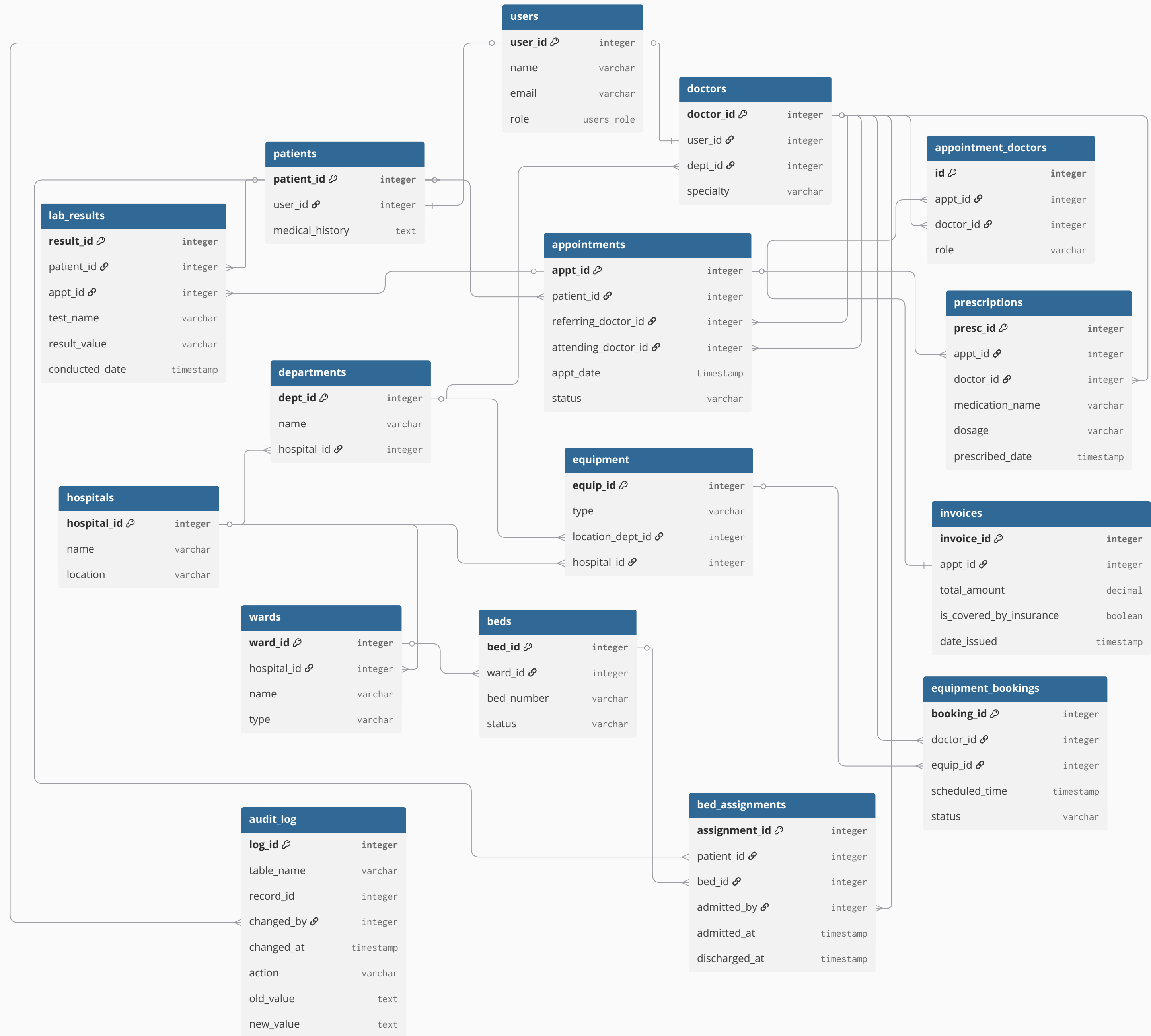
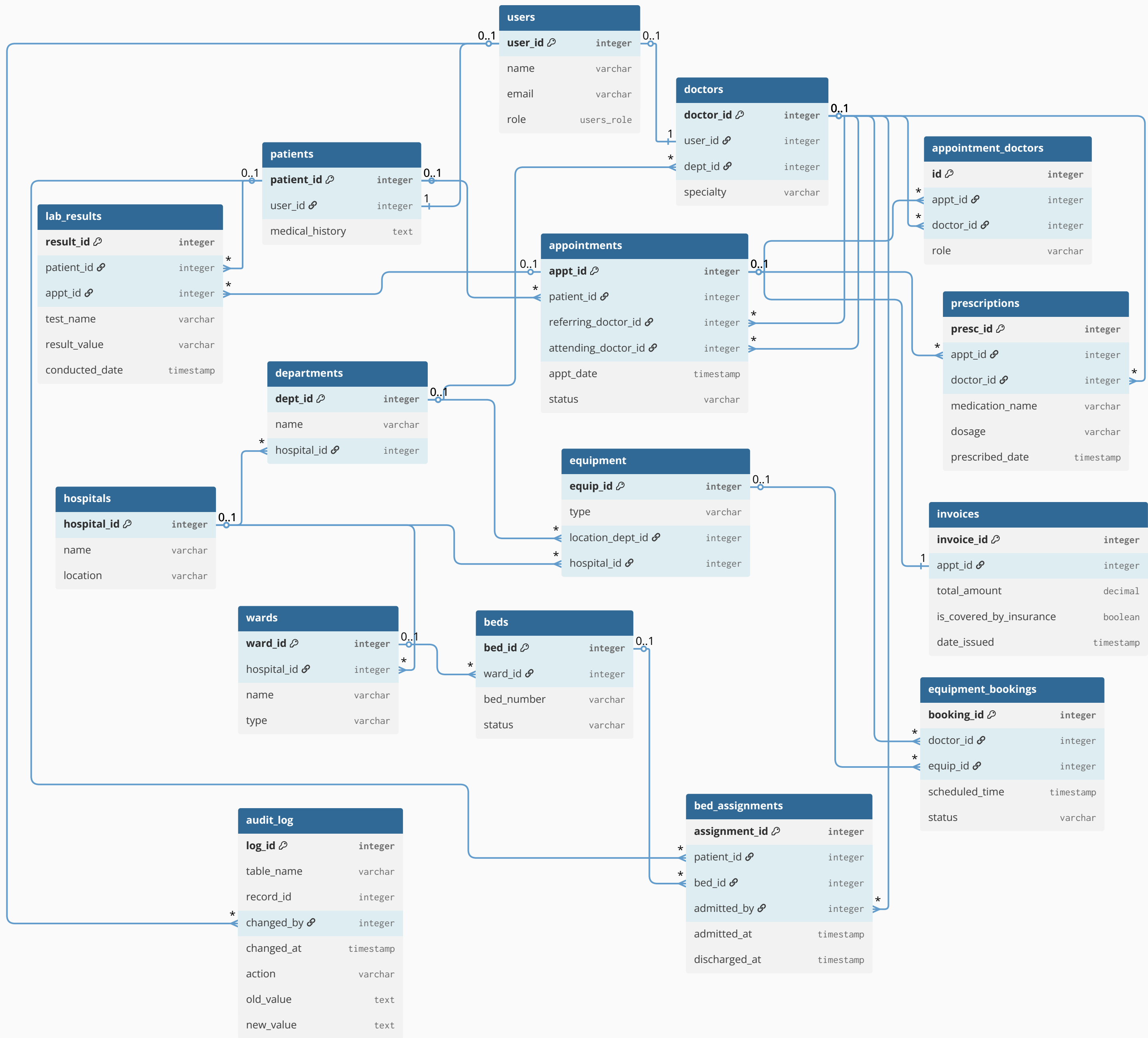






```
Table users {
  user_id integer [primary key]
  name varchar
  email varchar
  role users_role [note: "ENUM: 'patient', 'doctor', 'admin'"]
}
```

```
Table patients {
  patient_id integer [primary key]
  user_id integer [ref: - users.user_id]
  medical_history text
}
```

```
Table doctors {
  doctor_id integer [primary key]
  user_id integer [ref: - users.user_id]
  dept_id integer [ref: > departments.dept_id]
  specialty varchar
}
```

```
// Core Entities
Table hospitals {
  hospital_id integer [primary key]
  name varchar
  location varchar
}
```

```
Table departments {
  dept_id integer [primary key]
  name varchar
  hospital_id integer [ref: > hospitals.hospital_id]
}
```

```
// Resources
Table equipment {
  equip_id integer [primary key]
  type varchar // 'XRAY', 'MRI', 'ULTRASOUND'
  location_dept_id integer [ref: > departments.dept_id]
  hospital_id integer [ref: > hospitals.hospital_id]
}
```

```
Table equipment_bookings {
  booking_id integer [primary key]
  doctor_id integer [ref: > doctors.doctor_id]
  equip_id integer [ref: > equipment.equip_id]
  scheduled_time timestamp
  status varchar // 'scheduled', 'in_use', 'completed', 'cancelled'
}
```

```
// Patient Appointments
Table appointments {
  appt_id integer [primary key]
  patient_id integer [ref: > patients.patient_id]
  referring_doctor_id integer [ref: > doctors.doctor_id]
  attending_doctor_id integer [ref: > doctors.doctor_id]
  appt_date timestamp
  status varchar
}
```

```
// Many-to-many: multiple doctors per appointment
Table appointment_doctors {
  id integer [primary key]
  appt_id integer [ref: > appointments.appt_id]
  doctor_id integer [ref: > doctors.doctor_id]
  role varchar // 'surgeon', 'anesthesiologist', 'resident', etc.
}
```

```
Table prescriptions {
  presc_id integer [primary key]
  appt_id integer [ref: > appointments.appt_id]
  doctor_id integer [ref: > doctors.doctor_id]
  medication_name varchar
  dosage varchar
  prescribed_date timestamp
}
```

```
Table lab_results {
  result_id integer [primary key]
  patient_id integer [ref: > patients.patient_id]
  appt_id integer [ref: > appointments.appt_id]
  test_name varchar
  result_value varchar
  conducted_date timestamp
}
```

```
// Billing
Table invoices {
  invoice_id integer [primary key]
  appt_id integer [ref: - appointments.appt_id]
  total_amount decimal
  is_covered_by_insurance boolean
  date_issued timestamp
}
```

```
// Inpatient / Bed Management
Table wards {
  ward_id integer [primary key]
  hospital_id integer [ref: > hospitals.hospital_id]
  name varchar
  type varchar // 'ICU', 'general', 'maternity', etc.
}
```

```
Table beds {
  bed_id integer [primary key]
  ward_id integer [ref: > wards.ward_id]
  bed_number varchar
  status varchar // 'available', 'occupied', 'maintenance'
}
```

```
Table bed_assignments {
  assignment_id integer [primary key]
  patient_id integer [ref: > patients.patient_id]
  bed_id integer [ref: > beds.bed_id]
  admitted_by integer [ref: > doctors.doctor_id]
  admitted_at timestamp
  discharged_at timestamp [note: "NULL if still admitted"]
}
```

```
// Audit
Table audit_log {
  log_id integer [primary key]
  table_name varchar
  record_id integer
  changed_by integer [ref: > users.user_id]
  changed_at timestamp
  action varchar // 'INSERT', 'UPDATE', 'DELETE'
  old_value text // JSON snapshot
  new_value text // JSON snapshot
}
```